



Laboratorio: Introducción a Amazon Aurora (Interfaz Actualizada)

Atributo	Detalle
Dificultad	Intermedio
Tiempo Estimado	40 minutos
Servicios Principales	Amazon Aurora (MySQL Compatible), Amazon EC2, Amazon RDS



Resumen y Objetivos

Este laboratorio técnico se enfoca en el despliegue de **Amazon Aurora**, un motor de base de datos relacional nativo de la nube que ofrece alto rendimiento y disponibilidad. El objetivo es que domines el aprovisionamiento de clústeres Aurora y la gestión de conectividad segura desde capas de cómputo (EC2).

Al finalizar, habrás logrado:

- Aprovisionar un clúster de Amazon Aurora mediante el flujo moderno de la consola.
- Gestionar el acceso de red mediante Security Groups y Subnet Groups.
- Establecer comunicación segura mediante el protocolo MySQL/MariaDB.
- Validar la persistencia de datos y ejecución de consultas DDL/DML.



Análisis del Escenario

El cliente solicita una infraestructura de base de datos que soporte cargas de trabajo críticas con una latencia mínima. Se ha diagnosticado que Amazon Aurora es la solución ideal debido a su arquitectura de almacenamiento distribuido. La seguridad es una prioridad, por lo que el despliegue debe realizarse en subredes privadas dentro de una **VPC (LabVPC)**, permitiendo el acceso únicamente a través de un **Host de Comandos** autorizado, cumpliendo con el principio de defensa en profundidad.



Arquitectura de la Solución

1. **Capa de Datos:** Clúster de Amazon Aurora (1 instancia primaria) en subredes privadas.
2. **Capa de Gestión:** Instancia EC2 (Command Host) con el cliente MariaDB instalado.
3. **Red:** LabVPC con aislamiento de tráfico mediante Security Groups (puerto 3306).



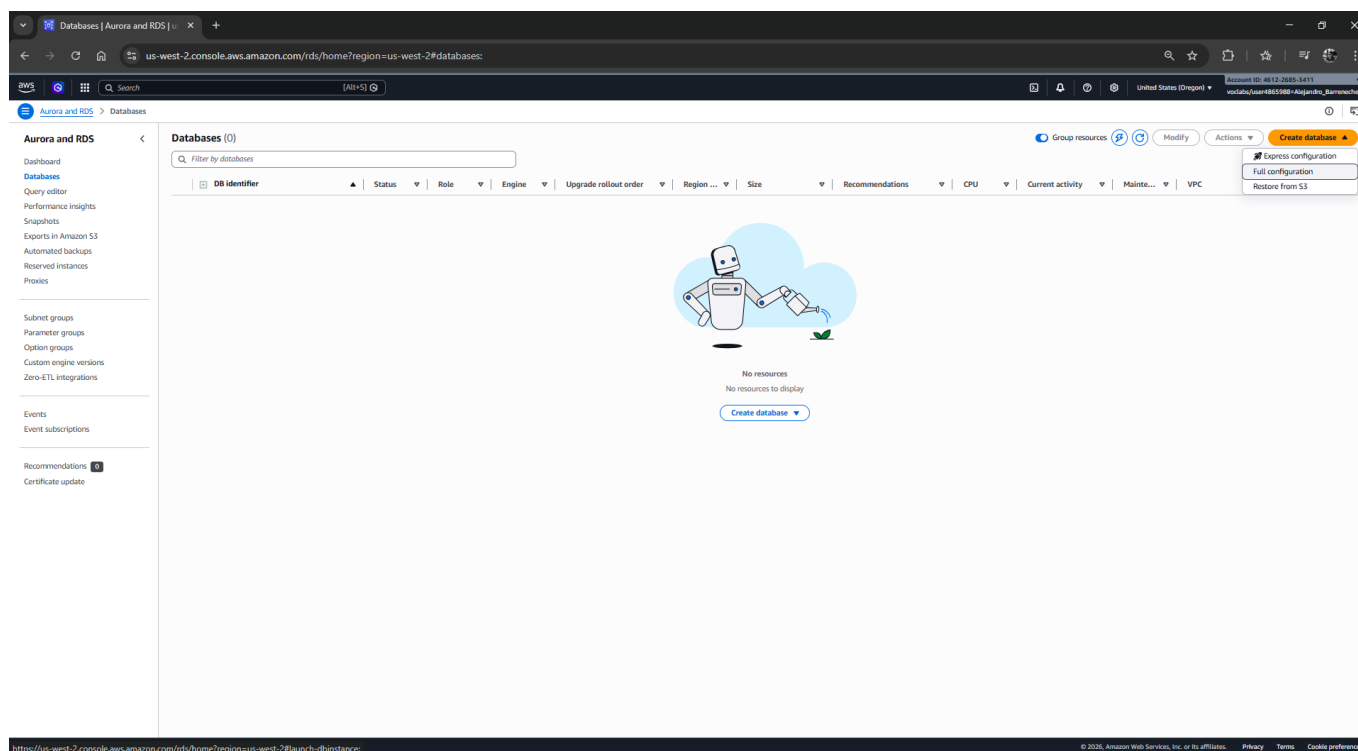
Desarrollo

Tarea 1: Creación del Clúster de Amazon Aurora

Sigue estos pasos adaptados al nuevo flujo de creación de base de datos de AWS:

1. Navega al servicio **RDS** desde la consola de administración.

- En el panel de navegación izquierdo, selecciona **Databases** y haz clic en el botón naranja **Create database**.



- En **Choose a database creation method**, asegúrate de seleccionar **Standard create**.
- En **Engine options**:
 - Engine type**: Selecciona **Amazon Aurora**.
 - Edition**: Elige **Amazon Aurora MySQL-Compatible Edition**.
 - Engine version**: Deja la versión por defecto (recomendada para Aurora MySQL 3 o superior).
- En **Templates**, selecciona **Dev/Test**.
- En la sección **Settings**:
 - DB cluster identifier**: Escribe **aurora**.
 - Master username**: Escribe **admin**.
 - Credentials management**: Selecciona **Self managed** y escribe **admin123** en el campo **Master password** y confírmalo.
- En **Instance configuration**:
 - DB instance class**: Selecciona **Burstable classes (includes t classes)** y elige **db.t3.medium**.
- En **Availability & durability**: Selecciona **Don't create an Aurora Replica**.
- En **Connectivity** (Sección crítica para el acceso):
 - Compute resource**: Selecciona **Don't connect to an EC2 compute resource** (la conectividad se gestionará manualmente).
 - Virtual private cloud (VPC)**: Selecciona **LabVPC**.
 - DB subnet group**: Elige **dbsubnetgroup**.
 - Public access**: Selecciona **No**.
 - VPC security group**: Selecciona **Choose existing**. **Importante**: Haz clic en la "X" del grupo **default** y selecciona **DBSecurityGroup**.
- Desliza hasta el final y expande **Additional configuration**:
 - Initial database name**: Escribe **world**.
 - Backup**: Desmarca **Enable automated backups**.
 - Encryption**: Desmarca **Enable encryption** (para fines didácticos del laboratorio).

- **Monitoring:** Desmarca **Enable Enhanced monitoring.**
- **Maintenance:** Desmarca **Enable auto minor version upgrade.**

11. Haz clic en **Create database.** El estado pasará de **Creating** a **Available** en unos minutos.

The screenshot shows the 'Create database' page in the AWS Management Console. The 'Engine type' section has 'Aurora (MySQL Compatible)' selected. The 'Choose a database creation method' section has 'Full configuration' selected. The 'Templates' section has 'Dev/Test' selected. The 'Cluster scalability type' section has 'Provisioned' selected. The 'Type of provisioned configuration' section has 'Burstable classes (includes 1 classes)' selected. The 'db.t3.medium' instance type is selected.

The screenshot shows the 'Settings' section of the 'Create database' page. The 'Engine version' is set to 'Aurora MySQL 3.10.3 (compatible with MySQL 8.0.42)'. The 'DB cluster identifier' is 'aurora'. The 'Master username' is 'admin'. The 'Credentials management' section has 'Self managed' selected. The 'Master password' is '*****'. The 'Confirm master password' is '*****'. The 'Additional credentials settings' section is expanded, showing 'Database authentication' and 'IAM database authentication' options.

Create database | Aurora and RDS

us-west-2.console.aws.amazon.com/rds/home?region=us-west-2#launch-dbinstance:

Search

[Alt+S] Ask Amazon Q

Aurora and RDS

Databases

Create database

Connectivity

Compute resource

Don't connect to an EC2 compute resource

Network type

Virtual private cloud (VPC)

DB subnet group

Public access

VPC security group (firewall)

Existing VPC security groups

Availability Zone

RDS Proxy

Certificate authority - optional

Connect to an EC2 compute resource

Dual-stack mode

LabVPC (vpc-01b48641c26897db3)

After a database is created, you can't change its VPC.

dbsubnetgroup

Yes

No

Choose existing

Create new

Choose one or more options

DBSecurityGroup

No preference

Create an RDS Proxy

Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

Log exports

IAM role

Devops Guru

Additional configuration

Initial database name

DB cluster parameter group

DB parameter group

Fallover priority

Backup

Backup retention period

Backup window

Backup tags

Encryption key

Backtrack

Maintenance

world

default:aurora-mysql8.0

default:aurora-mysql8.0

No preference

7

Choose a window

No preference

Copy tags to automated backup

AWS owned KMS key (SSE-RDS)

Enable Backtrack

Enable auto minor version upgrade

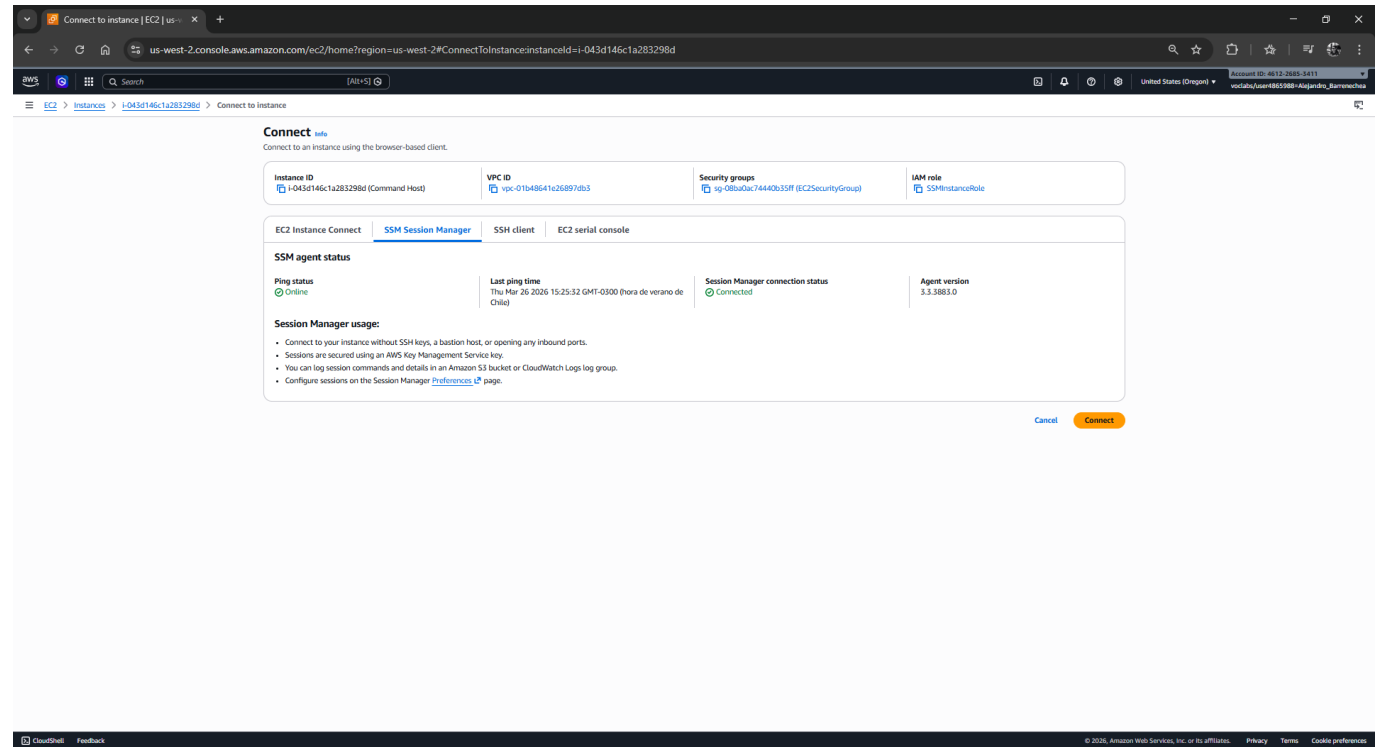
The screenshot shows the AWS Aurora and RDS console for the 'aurora' database instance. The instance is named 'aurora' and is in the 'Available' state. It is a Regional dual-AZ instance with 1 instance of Aurora MySQL in the us-west-2 region. The instance is connected to a VPC named 'vpc-01b4...' and has a security group named 'sg-01b4...'. The instance is using the 'aurora-instance-1' as the writer instance. The console shows the 'Connectivity & security' tab, which includes options for 'Connect using' (Code snippets, CloudShell, Endpoints), 'Internet access gateway' (Disabled), 'IAM Authentication' (Disabled), and 'Connection steps' (1. curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem; 2. mysql -h aurora.cluster-ckiesapqk.us-west-2.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=/global-bundle.pem). The 'Connected compute resources' section shows no connected resources.

Tarea 2: Conexión al Command Host

Establece la conexión al host administrativo para interactuar con la base de datos:

1. Navega al servicio **EC2** y selecciona **Instances**.
2. Marca la casilla del **Command Host**, haz clic en **Connect** y selecciona la pestaña **Session Manager**.
3. Haz clic en el botón naranja **Connect**.

The screenshot shows the AWS EC2 console for the 'Instances' page. The instance 'i-043d146c1a283298d' (Command Host) is selected and is in the 'Running' state. The console shows the 'Details' tab, which includes information about the instance's public and private IP addresses, DNS names, VPC ID, Subnet ID, and IAM role. The instance is running on the 't3.medium' instance type in the 'us-west-2a' Availability Zone. The console also shows the 'Status and alarms' tab, which includes the 'Instance state' and 'Alarm status'.

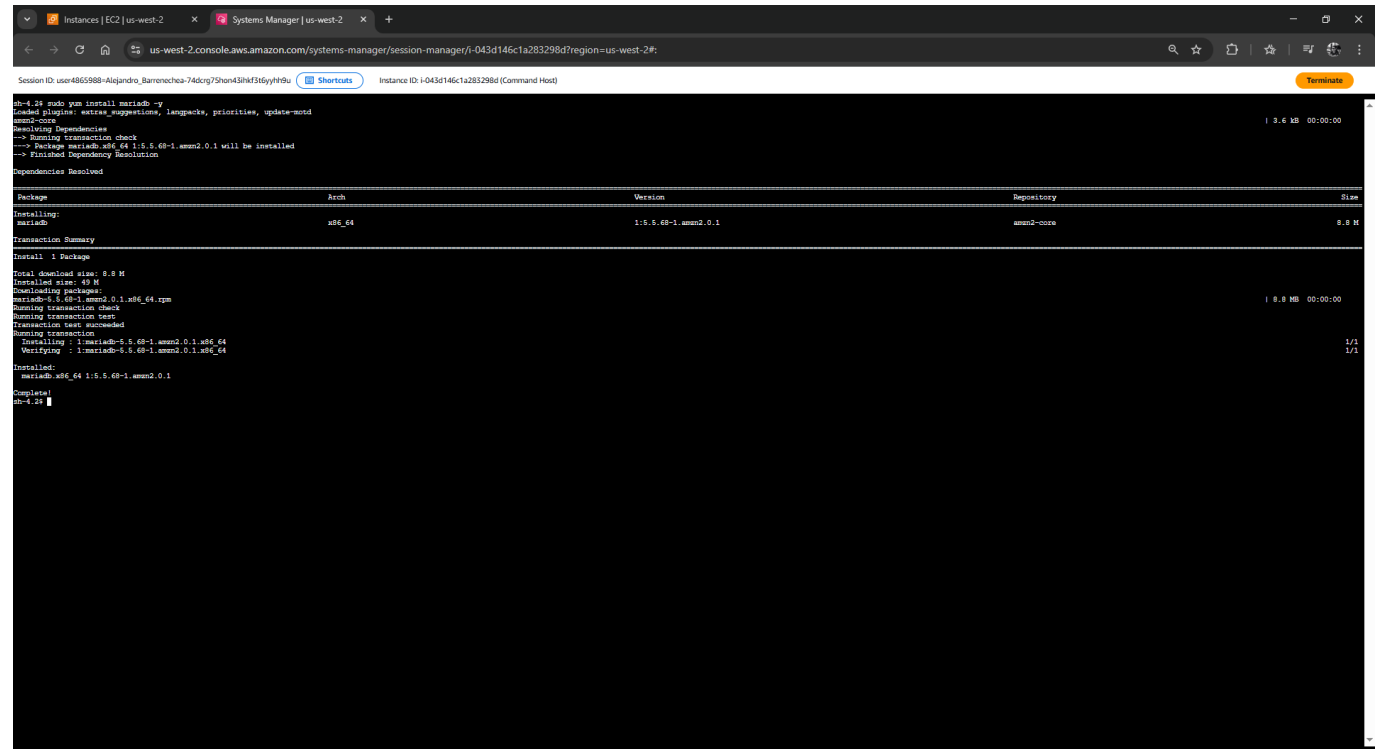


Tarea 3: Configuración del Cliente e Interacción con Aurora

Configura el cliente y obtén el punto de enlace de la base de datos:

1. En la terminal del Command Host, instala el cliente SQL:

```
sudo yum install mariadb -y
```



- Regresa a la consola de RDS, haz clic en el nombre de tu base de datos **aurora**.
- En la pestaña **Connectivity & security**, copia el **Endpoint** bajo la sección **Endpoint & port** (el tipo debe ser *Writer instance*).

The screenshot shows the AWS Aurora console for a database instance named 'aurora'. The 'Connectivity & security' tab is active, displaying the 'Endpoints' section. A table lists the endpoints:

Endpoint	Status	Type	Port
aurora.cluster-ck8iesapag-us-west-2.rds.amazonaws.com	Available	Writer	3306
aurora.cluster-ro-d8iesapag-us-west-2.rds.amazonaws.com	Available	Reader	3306

- Conéctate a Aurora usando el siguiente comando (reemplaza el endpoint):

```
mysql -u admin --password='admin123' -h <TU_ENDPOINT_COPIADO>
```

The screenshot shows a terminal window on an Amazon EC2 instance. The command `yum install mariadb -y` has been executed, and the output shows the successful installation of MariaDB 10.5.6-1-aurora2.0.1. The terminal output includes the following details:

```

dnf-4.29 sudo yum install mariadb -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-mo
mariadb-core
--> Running transaction check
--> Package mariadb.x86_64 1:10.5.6-1-aurora2.0.1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

Package Arch Version Repository Size
Installing:
mariadb x86_64 1:10.5.6-1-aurora2.0.1 mariadb-core 8.8 M

Transaction Summary
Install 1 Package
Total download size: 8.8 M
Installed size: 49 M
Downloading packages:
mariadb-10.5.6-1-aurora2.0.1.x86_64.rpm
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
Installing : 1:mariadb-10.5.6-1-aurora2.0.1.x86_64
Verifying : 1:mariadb-10.5.6-1-aurora2.0.1.x86_64
Installed:
mariadb.x86_64 1:10.5.6-1-aurora2.0.1

Complete!
dnf-4.29 mysql -u admin --password='admin123' -h aurora.cluster-ck8iesapag-us-west-2.rds.amazonaws.com
Welcome to the MariaDB monitor. Commands end with ; or \q
Your MySQL connection id is 43
Server version: 10.5.6-1-aurora2.0.1
Copyright (c) 2000, 2019, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql [(none)]>

```

Tarea 4: Gestión de Tablas y Consultas DDL/DML

Ejecuta la lógica de base de datos una vez conectado:

1. Accede a la base de datos inicial:

```
USE world;
```

2. Crea la tabla de países:

```
CREATE TABLE country (  
  Code CHAR(3) PRIMARY KEY,  
  Name CHAR(52) NOT NULL,  
  Continent ENUM('Asia','Europe','North  
America','Africa','Oceania','Antarctica','South America') DEFAULT 'Asia',  
  Population INT(11) DEFAULT 0,  
  GNP FLOAT(10,2) DEFAULT 0.00  
);
```

3. Inserta registros para validación:

```
INSERT INTO country (Code, Name, Continent, Population, GNP)  
VALUES ('IRL','Ireland','Europe',3775100,75921.00),  
      ('AUS','Australia','Oceania',18886000,351182.00);
```

4. Ejecuta una consulta de filtrado:

```
SELECT Name, Population FROM country WHERE GNP > 50000;
```


aurora - Database Details | Au...

Systems Manager | us-west-2

us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-043d146c1a283298d?region=us-west-2#

Session ID: user4865988-Alejandro_Baronechea-74dc972hon43hnf5tgyh9u

Shortcuts

Instance ID: i-043d146c1a283298d (Command Host)

Terminate

sh-4.2\$ sudo yum install mariadb -y

Loaded plugins: extras_suggestions, langpacks, priorities, update-selinux

Resolving Dependencies

Running transaction check

Package mariadb.x86_64 1:5.5.68-1.amzn2.0.1 will be installed

Finished Dependency Resolution

Dependencies Resolved

Package	Arch	Version	Repository	Size
Installing: mariadb	x86_64	1:5.5.68-1.amzn2.0.1	amzn2-core	8.8 M

Transaction Summary

Install 1 Package

Total download size: 8.8 M

Installed size: 49 M

Downloading packages:

mariadb-5.5.68-1.amzn2.0.1.x86_64.rpm

Running transaction check

Running transaction test

Transaction test succeeded

Running transaction

Installing : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64

Verifying : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64

Installed:

mariadb.x86_64 1:5.5.68-1.amzn2.0.1

Complete!

sh-4.2\$ mysql -u admin --password='admin123' -h aurora.cluster-ckless0pqq.us-west-2.rds.amazonaws.com

Welcome to the MariaDB monitor. Commands end with ; or \g

Your MySQL connection id is 63

Server version: 8.0.42 6252a39a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h;' for help. Type '\c;' to clear the current input statement.

MySQL [(none)]>

aurora - Database Details | Au...

Systems Manager | us-west-2

us-west-2.console.aws.amazon.com/systems-manager/session-manager/i-043d146c1a283298d?region=us-west-2#

Session ID: user4865988-Alejandro_Baronechea-74dc972hon43hnf5tgyh9u

Shortcuts

Instance ID: i-043d146c1a283298d (Command Host)

Terminate

sh-4.2\$ sudo yum install mariadb -y

Loaded plugins: extras_suggestions, langpacks, priorities, update-selinux

Resolving Dependencies

Running transaction check

Package mariadb.x86_64 1:5.5.68-1.amzn2.0.1 will be installed

Finished Dependency Resolution

Dependencies Resolved

Package	Arch	Version	Repository	Size
Installing: mariadb	x86_64	1:5.5.68-1.amzn2.0.1	amzn2-core	8.8 M

Transaction Summary

Install 1 Package

Total download size: 8.8 M

Installed size: 49 M

Downloading packages:

mariadb-5.5.68-1.amzn2.0.1.x86_64.rpm

Running transaction check

Running transaction test

Transaction test succeeded

Running transaction

Installing : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64

Verifying : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64

Installed:

mariadb.x86_64 1:5.5.68-1.amzn2.0.1

Complete!

sh-4.2\$ mysql -u admin --password='admin123' -h aurora.cluster-ckless0pqq.us-west-2.rds.amazonaws.com

Welcome to the MariaDB monitor. Commands end with ; or \g

Your MySQL connection id is 63

Server version: 8.0.42 6252a39a

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h;' for help. Type '\c;' to clear the current input statement.

MySQL [(none)]> USE world;

Database changed

MySQL [(none)]> CREATE TABLE country (

-> CODE CHAR(10) PRIMARY KEY,

-> NAME CHAR(50) NOT NULL,

-> CONTINENT ENUM('Asia','Europe','North America','Africa','Oceania','Antarctica','South America') DEFAULT 'Asia',

-> POPULATION INT(11) DEFAULT 0,

-> GNP FLOAT(10,2) DEFAULT 0.00

->);

Query OK, 0 rows affected, 2 warnings (0.02 sec)

MySQL [(none)]> INSERT INTO country (Code, Name, Continent, Population, GNP)

-> VALUES ('USA', 'United States', 'North America', 277100000, 18551.00),

-> ('AUS', 'Australia', 'Oceania', 18880000, 35112.00);

Query OK, 2 rows affected (0.00 sec)

Records: 2 Duplicates: 0 Warnings: 0

MySQL [(none)]> SELECT Name, Population FROM country WHERE GNP > 50000;

+-----+-----+

| Name | Population |

+-----+-----+

| Australia | 18880000 |

| Ireland | 277100 |

+-----+-----+

2 rows in set (0.00 sec)

MySQL [(none)]>

💡 Respuestas Analíticas y Diagnóstico Técnico

- ¿Cuál es la diferencia entre el "Cluster Endpoint" y el "Reader Endpoint" en la nueva interfaz? La consola moderna diferencia claramente estos puntos de enlace. El **Cluster Endpoint** (Writer) siempre dirigirá las conexiones a la instancia maestra para operaciones de escritura. El **Reader Endpoint** realiza un balanceo de carga automático entre todas las réplicas de lectura disponibles, optimizando el rendimiento de las consultas **SELECT** sin afectar al nodo principal.

- **¿Por qué el cliente MariaDB es la opción preferida para Aurora MySQL?** Debido a que MariaDB es un *fork* compatible de MySQL, sus herramientas de CLI ofrecen una paridad total con Aurora MySQL. Es una opción ligera que no requiere licencias adicionales y está optimizada para repositorios de Amazon Linux 2/2023.
- **¿Qué impacto tiene el "Public access: No" en la seguridad del clúster?** Al deshabilitar el acceso público, AWS garantiza que la base de datos no tenga una interfaz de red accesible desde internet. Esto mitiga ataques de fuerza bruta externos y asegura que el tráfico sea exclusivamente interno a través de la red de AWS, validado por las reglas de entrada del Security Group.

¡Felicidades! Has completado el laboratorio de Amazon Aurora utilizando la interfaz de consola más reciente y siguiendo estándares de arquitectura segura.